

Rajarshi Chakraborty

Portfolio | GitHub | LinkedIn | LeetCode

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EDUCATION

- Techno Main Salt Lake** Kolkata, India
Bachelor of Technology - Computer Science and Engineering; GPA: 8.01 July 2023 – Present
Courses: Operating Systems, Data Structures and Algorithms, Analysis of Algorithms, Database Management Systems, Computer Networks, Object-Oriented Programming, Theory of Computation, Discrete Mathematics, Software Engineering, Computer Organization and Architecture, Cryptography and Network Security

SKILLS

- Languages:** Python, C/C++, Java, JavaScript, TypeScript, SQL, Bash
- Frameworks:** React, Next.js, TailwindCSS, ShadCN, Node.js, Express.js, Scikit-learn, NLTK, SpaCy, TensorFlow, Keras
- Tools:** Git, Docker, PostgreSQL, MySQL, MongoDB, Prisma, Mongoose, Drizzle, Postman, CI/CD Pipelines
- Platforms:** Linux, Windows, AWS (EC2, S3), Supabase, Vercel
- Soft Skills:** Leadership, Technical Writing, Team Collaboration, Public Speaking, Time Management

EXPERIENCE

- Techno Main Salt Lake** Kolkata
Undergraduate Researcher 2026 – Present
 - Time-Series Modeling & Forecasting:** Developed **LSTM, ARIMA, and RNN-based models** for weather prediction using **time-series analysis** on real-world datasets, leveraging **statistical modeling** and **deep learning** techniques.
 - Production-Ready ML Systems:** Built **scalable, deployment-ready ML pipelines** using **Python, NumPy, Pandas, Scikit-learn, PyTorch, Node.js**, and time-series frameworks for end-to-end implementation.

PROJECTS

- SynapseUI – React UI Component Library** [[Demo](#) — [GitHub](#)] (Next.js, TypeScript, Tailwind CSS, Fumadocs, ShadCN CLI): Built a **production-ready, open-source UI component library** for React and Next.js applications, enabling developers to **copy-paste components** directly into their workspace or install via **ShadCN CLI**. Architected a **fully customizable component system** with **TypeScript** type safety, styled with **Tailwind CSS** and **ShadCN** primitives. Delivered comprehensive **documentation** powered by **Fumadocs**, with live component previews and CLI-based scaffolding. Tech: **Next.js, React, TypeScript, Tailwind CSS, ShadCN, Fumadocs, CLI Tooling** (2026)
- SocketPoll – Real-Time Polling App** [[Demo](#) — [GitHub](#)] (Full Stack, WebSockets, Event-Driven Architecture, React): Designed a **low-latency, event-driven polling platform** using **WebSocket (Socket.IO)** for **bidirectional real-time communication**, enabling instant **vote propagation** with zero page refresh. Built a **RESTful Express.js** backend with **room-based namespace isolation** and **in-memory state management** for concurrent sessions. Developed a **reactive Vite + React** frontend with **optimistic UI updates** and animated result visualization. Tech: **Vite, React, Express.js, Socket.IO, Node.js, REST APIs, WebSocket Protocol** (2026)
- LinkedUp – LinkedIn Post Generator** [[Demo](#) — [GitHub](#)] (Generative AI, LLM Orchestration, NLP, Streamlit): Engineered an **LLM-powered content generation pipeline** using **LangChain** for **prompt chaining**, **retrieval-augmented generation (RAG)**, and **output parsing** to synthesize context-aware **LinkedIn posts** tailored to tone, domain, and audience. Integrated **few-shot prompting** and **memory buffers** for multi-turn coherence with real-time token streaming. Tech: **Python, LangChain, OpenAI API, Streamlit, Prompt Engineering, Few-Shot Learning** (2026)
- Mini-Tailwind CSS Engine** [[Demo](#) — [GitHub](#)] (Frontend Engineering, CSS Runtime, DOM Manipulation, CDN Architecture): Architected a **zero-dependency, utility-first CSS runtime engine** inspired by Tailwind CSS with a **custom class-parsing pipeline** and **real-time DOM MutationObserver** for dynamic style injection without a build step. Achieved sub-millisecond **style reconciliation** via an in-memory **stylesheet diffing algorithm**. Tech: **Vanilla JavaScript, MutationObserver API, CSS Object Model (CSSOM), CDN Architecture** (2026)

HONORS & CODING PLATFORMS

- Solved **580+** problems on [LeetCode](#) with **10+** contest participations; strong **data structures and algorithms** proficiency
- Solved **520+** problems on [GeeksforGeeks](#) with a **1700+** coding score; consistent **competitive programming** performance
- Solved **120+** problems across [Codeforces](#) and [MasterJi](#); strong **algorithmic thinking** and **optimization** skills

VOLUNTEER EXPERIENCE

- Co-Head & Technical Content Writer – SAMARTH, TMSL** Kolkata, India
2024 – Present
 - Leadership & Execution:** Led the **IGNITE division**, driving **program management** and **community growth**. Organized **Eduathon** and **Safalya**, ensuring end-to-end execution and strong **stakeholder coordination**.
 - Content & Impact:** Created **technical content** and contributed to **content strategy, documentation**, and **outreach**, improving **user engagement** using **data-driven insights**.